

Brighten Hayama

brighyama@gmail.com | [LinkedIn](#) | [Personal Website](#)

EDUCATION

M.S. in Computer Science

Sep 2025 – Present

University of California, San Diego

La Jolla, CA

B.S. in Mathematics and Computer Science, Minor in Data Science

Sep 2021 – Jun 2025

University of California, San Diego

La Jolla, CA

- Cum Laude – GPA: 3.91/4.00

EXPERIENCE

Undergraduate Instructional Assistant

Sep 2023 – Jun 2025

University of California, San Diego

La Jolla, CA

- Collaborated with professors to teach and reinforce key concepts in algorithm analysis, probability, and machine learning for courses including Mathematics for Algorithms/Systems, Theoretical Foundations of Data Science, Supervised Machine Learning, Optimization Methods for Data Science, and Stochastic Processes, improving student comprehension and engagement
- Led 100+ office hours and review sessions for groups of up to 15 students, clarifying algorithm runtimes, combinatorics, graph theory, optimization, regression, and classification to strengthen student problem-solving

Software Engineer Intern

Jun 2024 – Aug 2024

San Diego Supercomputer Center

La Jolla, CA

- Collaborated in a 6-person Agile team to design and develop a Yelp-inspired social web app for university students, integrating a responsive **React** frontend with an **Express.js** and **Firebase** backend to promote affordable and accessible local experiences
- Implemented friends features, interactive post and comment systems, and a custom review submission form, enabling students to share and discover nearby restaurants and locations through peer recommendations

PROJECTS

Machine Learning Interpretability Research

Jan 2024 – Present

Supervised by Prof. Sanjoy Dasgupta

- Explored feature representations in Convolutional Neural Networks using **PyTorch**, **Matplotlib**, and **scikit-learn**, computing statistics across network layers and refining code to streamline computation-heavy experiments through serialization and modular design
- Reviewed literature on Interpretable Machine Learning and presented **OpenAI's CLIP** model, effectively communicating its vision-language alignment mechanisms and insights to the research group

Type Inference Implementation

Mar 2025 – Jun 2025

Academic Project

- Implemented **Algorithm W** for type inference in 'Nano', a miniature **Haskell**-based language
- Designed type environment, unification, and substitution logic according to **Hindley-Milner** type systems
- Collaborated in a 4-person team to integrate independent Nano modules (pattern matching, parser, REPL)

NBA Player Interaction Network Analysis

Jan 2025 – Mar 2025

Academic Project

- Developed an evolving NBA player assist network from event-based **NBA API** data across multiple seasons
- Analyzed **centrality**, **community density**, and other metrics to quantify team chemistry and performance
- Visualized historical trends in player influence and team dynamics using **Python**, **NetworkX**, and **Matplotlib**

Weather Forecast Web App

Sep 2022 – Oct 2022

Personal Project

- Developed a weather web application using **React.js** that displays up-to-date hourly and weekly weather forecast
- Used **OpenWeatherMap API** to fetch real-time weather forecast data for any city in the world

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, HTML, CSS, SQL

Libraries: PyTorch, scikit-learn, pandas, NumPy, Matplotlib

Tools & Frameworks: PySpark, React.js, Express.js, Firebase, Git